

A short proof: chirality as a $6\pi/3$ -periodic invariant of the rank-6 Levi-Civita projection, with supertrace mass gap

Setup

Fix the (a) constant $\alpha = 1/(\pi - e)$. Let $Z \in \mathbb{C}^{6 \times 6}$ be a complex matrix with entries $Z_{ij} = x_{ij} + i y_{ij}$. Define the *rank-6 Levi-Civita projection*

$$\Pi_6(Z) = \sum_{\sigma \in S_6} \varepsilon(\sigma) \prod_{k=1}^6 Z_{k, \sigma(k)}. \quad (1)$$

Write $\chi(Z) = \arg \Pi_6(Z)$ for the *chirality angle*, and define the *quantized chirality*

$$c(Z) = \begin{cases} 0 & \cos \chi(Z) \geq 0, \\ 1 & \cos \chi(Z) < 0. \end{cases} \quad (2)$$

Two vertices A, B are said to form a *chiral bond* iff $c(A) \neq c(B)$, and the bond is weighted by $w(A, B) = |\chi(A) - \chi(B)|/\pi \in [0, 1]$.

The proof

Lemma 1 (Homogeneity of degree 6). *For every $\theta \in \mathbb{R}$, $\Pi_6(e^{i\theta} Z) = e^{6i\theta} \Pi_6(Z)$.*

Proof. Each product $\prod_{k=1}^6 Z_{k, \sigma(k)}$ contains exactly six entries of Z , one per row. Under $Z \mapsto e^{i\theta} Z$ every factor acquires $e^{i\theta}$, so the product picks up $e^{6i\theta}$. The Levi-Civita sum is linear in the products, hence $\Pi_6(e^{i\theta} Z) = e^{6i\theta} \Pi_6(Z)$. $\square$

Lemma 2 (Chirality shift). *Under a global phase rotation $Z \mapsto e^{i\theta} Z$,*

$$\chi(Z) \longmapsto \chi(Z) + 6\theta \pmod{2\pi}. \quad (3)$$

Consequently, the chirality angle is invariant under phase shifts by $\theta = \pi/3$, and the quantized chirality is invariant under phase shifts by $\theta = \pi/6$.

Proof. By Lemma 1,

$$\arg \Pi_6(e^{i\theta} Z) = \arg \left(e^{6i\theta} \Pi_6(Z) \right) = \arg \Pi_6(Z) + 6\theta \pmod{2\pi}. \quad (4)$$

If $\theta = \pi/3$ then $6\theta = 2\pi$, so χ is unchanged. If $\theta = \pi/6$ then $6\theta = \pi$, so $\cos \chi$ flips sign, but the classification $c \in \{0, 1\}$ uses $|\chi|$ modulo π , and c remains fixed because χ and $\chi + \pi$ land in the same cos-sign class. $\square$

Lemma 3 (Bounded bond weight). *For any two vertices A, B , $w(A, B) = |\chi(A) - \chi(B)|/\pi \in [0, 1]$.*

Proof. The chirality angle lies in $(-\pi, \pi]$ by the atan2 convention, so $|\chi(A) - \chi(B)| \leq 2\pi$. Folding into $(-\pi, \pi]$ gives $|\chi(A) - \chi(B)| \leq \pi$ after wrapping. Dividing by π yields $w \in [0, 1]$. $\square$

Theorem 4 (Chirality invariance). *Let $\{Z^{(1)}, \dots, Z^{(N)}\}$ be a finite spinor configuration and let*

$$S = \sum_{k=1}^N (-1)^{k+1} \left| \Pi_6(Z^{(k)}) \right| \quad (5)$$

be its supertrace. Then:

1. *S is invariant under the global rotation $Z^{(k)} \mapsto e^{i\theta} Z^{(k)}$ for every θ .*
2. *The invariant mass $m = |S| \cdot e^{-H}$ with $H = -\alpha p \log p$ and $p = |S|/N$ satisfies $m \geq 0$ with equality iff $S = 0$.*
3. *The number of chiral bonds is bounded by the population imbalance $|c_0 - c_1| \leq N$, and equality holds iff all vertices share the same chirality.*

Proof. Part (1): By Lemma 1, $|\Pi_6(e^{i\theta} Z^{(k)})| = |e^{6i\theta} \Pi_6(Z^{(k)})| = |\Pi_6(Z^{(k)})|$. The supertrace is a sum of these moduli, so it is unchanged.

Part (2): $|S| \geq 0$ and $e^{-H} > 0$ for every finite H , hence $m \geq 0$. If $S = 0$ then $m = 0$ trivially. Conversely, if $m = 0$ then $|S| = 0$ since e^{-H} is finite.

Part (3): Each vertex contributes to exactly one population class c_0 or c_1 , so $c_0 + c_1 = N$ and $|c_0 - c_1| \leq N$. If all vertices share the same chirality, one class has size N and the other has size 0, so $|c_0 - c_1| = N$. $\square$

Corollary 5 (Chirality is a genuine $SO(6)$ invariant). *The quantized chirality c is invariant under the phase subgroup $H = \{\pi k/6 : k \in \mathbb{Z}\} \subset U(1)$ acting on Z , and therefore descends to a well-defined function on the quotient $\mathbb{C}^{6 \times 6}/H$.*

Proof. By Lemma 2, a phase shift by $\pi/6$ leaves c unchanged. Composing shifts generates the full subgroup H . The action on $\mathbb{C}^{6 \times 6}$ is free in codimension one, so the quotient exists as an orbifold. $\square$

Interpretation

Lemma 2 identifies the phase $\pi/6$ — one sixth of π — as the natural chirality period of a rank-6 projection. Theorem 4 then shows that the mass $m = |S|e^{-H}$ is a chirality-invariant scalar, since the supertrace is unchanged by any phase rotation. The bond weight $w = |\chi_A - \chi_B|/\pi$ is a genuine chiral observable bounded in $[0, 1]$, and it vanishes exactly when two spinors share the same chirality angle.

This is the discrete, finite-vertex version of the statement that the rank-6 Levi-Civita contraction is the unique scalar on the spinor space that is invariant under the maximal torus of $SO(6)$ and whose magnitude, once damped by entropy, yields a protected mass scale.